

ActInf GuestStream 060.1 ~ "Reflections and implications of a naïve attitude to architecture"

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hello and welcome to the active inference Institute this is actin guest stream number 60.1 on October 18th 20123 today should be a great and very exciting presentation and discussion Kristen Whittle is here and we'll be presenting on Reflections and implications of a naive attitude to architect cure there will be a presentation followed by a discussion so if you're watching live please feel free to write any comments in the chat otherwise thank you so much for joining and really looking forward to this work being shared thank you Daniel um it's uh it's a great pleasure being able to present to you and uh your audience um and just as we were maybe talking the journey towards uh arriving here to talk to you has been an very very interesting one and one that's really been undertaken since I was a young person so um it's it's thrilling and exciting to be able to talk about the work that we've been doing as a studio and kind of seeing where the uh touch points and um points of influence and interest to your you and your audience um come into this uh conversation about our sort of artistic work I guess so what I wanted to do is to um maybe explain what we're just about to see uh before showing it um what it is is a kind of an assemblage of work that's been undertaken since I was probably 22 um and and it it goes into a lot of practical uh work that um has now been completed um real large scale buildings Etc so it's kind of part theoretical part kind of trying to understand as a curious person where I'm at and trying to evolve a maybe a a strong systematic view on architecture because I guess the way I see it is that architecture is is a bit lost um and and in a sense there's an idea implicit in the work which is to kind of remind ourselves the purpose of architecture in a kind of pre-industrial form and then looking towards the future uh towards a very sophisticated future that we we have in front of us and trying to synchronize that with a very humanist substrate humanist philosophy um so um the reason why I called the talk um uh a kind of evocation about naive architect ET actually comes back to some readings that we've been doing uh that Carl Jung uh evolved out of um readings that he'd had uh from schilla the German uh writer and philosopher about uh ways in which artistic production occur so one

would have for instance one production of art step from an natural source almost which he referred to as naive art and or a naive approach to Art and the other which is what he referred to as a Sentimental view or a Sentimental approach which is more uh that you're standing away from nature and looking upon it and choosing which direction to evolve in an objectifying the direction so the first thing to say is that our work really stems from an extrapolation of nature through our work in a kind of intuitive instinctive manner um and then maybe post rationalization or or more theoretical Explorations after that but the the very basics of it come from a more intuitive approach and I think is very exciting um so the primary function is this kind of relationship ship between a human being and space so um having done a lot of work in health uh become very very interested in how the environment affects emotion um um and later on we'll sort of be talking about how free energy and coherence works with the brain and things like this but but the essential kind of exciting area is this relationship between a human being emotion feeling and space in the environment and so our work is I guess coming from a kind of slightly utopian perspective seeking to reconcile um the maybe lack of connection in the modern world to the physical environment that's this kind of reward Matrix that we that sits outside of our bodies um to generate an architecture which is full of meaning and nooks and hooks and uh alos if you like emotional alos which hook us into this kind of positive feedback loop so we see our extrapolation of our cognitive creative processes being something to do with how nature and feedback and cognition works at a fundamental level so I'm very interested in exploring this with you and your audience later on is how creativity is a sort of um manifestation of cognition full stop um so with architecture the interesting thing about architecture is it's really associated with other people and it requires you to be an empath so the big thing here is when you meet a client who's got a problem you're really trying to solve their problem and that their problem is manifold in different has different sort of Concepts with that but I'm specifically interested in the well-being aspect um of how they themselves or their cohort that's um living inside their buildings how they behave and how well they feel uh once a building's been built so architecture extraverts and I guess through the process of capturing the spirit of an environment that's hopefully regenerative um what we are now working on is embodying that regenerative philosophy into a kind of code which is a complex code uh interwoven um and made complex because it's fairly wide ranging and doesn't like being in the boundaries of strict philosophical guidelines if you know what I mean it kind of ranges across live data and and wanders around that Matrix kind of freely the process of embodiment is more to do with how the process of experimentation and scoring

works at an intuitive level in order to get very good uh architectural results happening after that so essentially interested in the Life journey and how people grow and get nurtured over time in their environments whether that's the intimate environment of their personal life their family lives or the institutional lives that they have um so without being kind of um institutionalizing I'm interested in the modularity if you like of people in the way that their cognition behaviors work at a sort of a neurobiological level I'm interested in how idiosyncrasy comes about within within that system and then I'm interested and how those idiosyncrasies can manifest into an idiosyncratic architecture to to basically create positive feedback loops um so somehow within this there's a correlation between how cognition Works which I see as the foundation to creativity and then how the making process works associated with that and perfecting the making process and then obviously what the lived experience of those um environments is and then how does that feedback work um to go back into cognition and creativity so um I guess some references here might be useful um the the work that smutz did you know at the same time as when Yung was working in basle um trying to evolve a complex holistic approach to the mind of body uh has been instrumental in the work that we do um so in a sense uh one takes that sort of expansive infinitely expansive kind of perspective of the universe being fed into us and then as emitting that universe that sort of overall large scale complex process that's always happening somehow our creative process is captured within that so I use that as a bit of a reference point um and I'd like to just I guess put a shout out here to to corbusier because um even though he he manifest a scientific and maybe um definitely institutional um and scientific approach to the modular the human modular his Focus was really on the body um his artistic maybe intuitive side was maybe interested in the mind but the systems and modernistic rationale that he produced in the 20th century had a massive effect on on cities globally and ultimately kind of fell over post brutalism into a kind of failure of architecture I see the work that we're doing is somehow an evolution of the modula um and a much more comple and um diverse manifestation of the modul so um I guess coming back to car young again um it would be necessary for me to say that uh if you go along with this story of naivity and processing nature that one of those processes is to do with the archetypal image and they're sort of using the archetypal image as some sort of foundation for the work so if if this sort of stream of Consciousness is available to you that's one of the latching points of the work and the question then becomes how do you utilize that as a sort of febrile substance to manifest new things so not necessarily reproducing memory um as an exact thing in and of itself it's more associated with the creative utility of the

archetype as a substrate to tap into and then applying different processes onto that and because of that let's say use of that substrate we're able to get into a kind of deep uh relationship or a deep conversation with the public so um I think the reason why this is kind of important at this moment in time is you know the question about data um what data informs architecture um typically it's been ergonomic fact-based information that's determined by spatial mathematics um an ergonomic programming of space like utility uh efficiency and those sorts of Concepts um I'm sort of more interested in the pre-industrial which is more to do with scale um the idea of rooms the idea of enclosure and and connect uh to other people um and and what sort of spatial configurations allow for that um so the question I have is really how do we use data in that sense you know um outside of the engineering based mathematical scientific modernistic uh platform for data so which I think caused in institutionalization which has been seen as a failure in architecture and failure in society so we're sort of we've never recovered really from the failure of the institutionalization of architecture so we're now at the point with large data and AI to question what gets assembled from here on in like and how do we evolve something that's got the Nuance complexity freedom of our organic cell in in a kind of artificialized system so I'm very interested in individuation both as a creative person but also creating environments that allow for other people to be individuated and for that allowing that ecosystem to dominate so I'm therefore interested in industrial experimentation or in experimentalism and and how um Mass customization can occur using um new manufacturing techniques like you know additive manufacturing for instance um and in a sense democratizing the design process to allow everyone to be an artist to be an architect but anyway um so to kind of unpack what we would do typically um is rather than go on a linear Journey which is a fact-based re reduced simplistic architectural Journey like $a + b = c$ we go on a circular kind of coiling Journey uh it's certainly moving forward but information's coming in from different Vantage points continually throughout the process and I guess we've we've ended up being very efficient at this um but it allows us to have kind of oppositional trajectories coming into the design Journey it allows us to see things objectively as well as subjectively using qualitative and quantitative information and Building Things and destroying things on a journey of creation so things die certain things live the living things get tested and might die and it evolves in that sort of way okay so um if you're looking at maybe the definitions of creativity um it would probably Center on how creative processes end up creating new things like um it's not so just speak about mimicking it's about creating new things that have a function and I guess from our perspective this coiling process is the

foundation to that um invention process um where somehow you yourself because you yourself have connected into that information process that by necessity what comes out of it um traps uh the exter external world into that internal world and gets affected by that Journey so it's sort of a trap uh for for for society um but the main objective is for um one to achieve what I would call coherence which is um which definitely comes back to free energy and active inference where you can essentially map and see the world moving forward in a coherent way latch onto it and for it to feed back meaning and reward into your life and therefore extend you know longevity and improve livability those sorts of things so um by sort of going in that direction what what we need to do is to kind of assess what The Well of knowledge uh Plus memory uh plus ambition what what sort of constitutes that domain so what we've done in our own way is to assemble uh trajectories and kind of taxonomy a working taxonomy that ranges between social and political principles to Scientific and artist and technological principles and art and Architectural principles and that essentially forms a kind of rosette which we've called deep learning and that's a kind of foundation that has existed for me since childhood and obviously is a dynamic assembly process so that's the kind of scaffold and then if you imagine taking that living scaffold to a project we then um create a dynamic relationship uh between that scaffold to a client and the projections and information that we get to and from a client as well as to from a site so a site would for instance constitute how where the sun moves how it moves around a site where the wind blows what's the Ecology of the site um and in a city what are its neighbors what's the social um backdrop to a site and it's also about people obviously and it's about monitoring and assessing people and how they are coming and going within within our sites um curtilage so you're sort of assembling that very Broad and complex picture that essentially forms that nugget of Truth which we've defined as feeling so we trust our feelings about about a sight now it sounds naive talking about feelings in architecture and there's no sort of um I don't believe there's any sort of academic construct that's been developed around feelings um it's been sort of chipped away in in naist principles Po in postmodernist theory but um the idea of a feeling is obviously if you're trusting your instincts is a very complex um model of of truth that's a truth to you and so if if you were able to form a um a layered understanding about how that feeling is constructed you're then able to uh I guess talk about it openly with your studio and with your clients and start utilizing it and testing it more pragmatically um so what we've done is to um is to then layer that out from initiation phases all the way through schematic delivery and post occupancy phases of projects so it's is so that feel concept is the Nugget that everything is

tested against and so when when we get to the end of the process uh we're measuring against what that original feeling was and asking question does it feel like what we felt like At the beginning how does it differ and how what do other people think about that feeling and I guess it's probably important for me to say the Curiosity that underpins all of this uh started because the work that was happening ended up being very well fitted to the client's prerogatives and I would have a situation quite commonly where a client would say to me how did you read my mind like how were you able to comprehend what our needs were when it there was a lot of unspoken truth about a project um how were you able to encapsulate what our desires were in a project how is that possible and or in the sense in the hospital why does the hospital feel right why does it feel good why is it restorative like how have you been able to achieve that and because of these questions that came at me I decided back in 2019 to study it to work out why it was happening what fed it what underpinned so that's kind of what why I'm here really talking to you Daniel um and I guess what I've realized is that um that the the perception model that I've just talked about in terms of feeling um it has obviously coroller or an association with memory and anticipation so projection and somehow in a very complex um neur neurological process we're capturing that data and then we're able to project it into architectural models so if you imagine cognition Pro like intuitive cognition projecting scenarios to help guide your body through space what I do as an architect is I actually make models of that projection and I then test those models in order to get the perfect model the thing that's the most coherent model for me and for the the situation so it's a situationally specific cerent model that's what I'm searching for and that's what I build at the end of the day so it's based on feedback so we would call that an embodied holistic code for architecture okay so it's based on the idea that we are all modular people in the sense that the foundation for us all our neurobiological foundations are common um before they become ideosyncratic so to speak um and I'm sure you'll correct me on all that um so anyway so but it's fair to say that we are all living beings right and we we have all of these um normal instinctive kind of primitively derived things that we do in life and this is what unifies us so as an architect it's important for me to know how these actions uh work because they actually underpin how the functioning capacity for building works but it's also true to say that in the work we are trying to reduce anxiety not invoke anxiety we're trying to create uh Harmony rather than invoke uh complexity to the to the point that it creates neur U pathologies we we're trying to Harmon harmonize our existence in cities specifically and in contemporary cities this is where we go drastic wrong because the human being is never at the center of the conversation uh with

regards to a development it's always a spreadsheet driven exercise and this is where we go wrong so our work begins with studying uh how this works how this kind of human environment works these are cting hollers uh art piece uh this kind of objectifying the social domain of a restaurant which is very evocative but as a basic process and this is again humble me with an iPhone what I try to do all the time every day is to document what I would call Nooks uh where people are settling and where people are seeking to settle in certain domains whether that's to socialize or to have the sort of intimate moments of uh self-reflection how do we move through cities so what I'm always doing is photographing um basically cities and people and that forms the foundation so I call that um and again this is a very unacademic terminology learn dwell no so I've I've worked out that my fascination with photographing these things stems from the fact that I I'm learning something from what I'm looking at and that's why I'm drawn to them and then also it's conflated with the idea that I want to live in that environment so there's there's something about the domain that's on photographing that combines and conflates learning something and living in somewhere that's going to be good for me so it's a compilation of those two things so our work's very much about the assemblage and collecting of that rationalized domain of live dwell no uh so we're always fascinated with how mattera for instance assembles itself so this is um an aerial photograph of a region of Northern Victoria a desert region of Northern Victoria which um defines this um it's an irrigation colony called mjur which um is a desert region that became an an agricultural agrarian society that was all to do with growing um food food for the city of Melbourne and but but if you look at it from on top you get to see the literal translation of those ecosystems being played out by the patterns of nature so it's the patterns of nature that we're interested in here we're interested in how um we sanctify our existence through symbolic constructs this again goes back to the archetypes we're interested in materials that are deep understanding about the kind of exacting expression of nature uh this is we have some fascinating trees in in Victoria in Australia and um I became fascinated in trees when I did The Royal Children's Hospital um there always a foundation pillar for me anyway um materials that maybe are transformative materials that are ambiguous uh especially drawn to those sorts of materials um materials that have maybe handcrafted and slightly Sublime complex process that's created them um materials that have used literal tree trunks in buildings that maybe have an unusually primitive quality to them as well as highly elaborate expressions of ress and thresholds that we see In classical architecture or humble um quiet domains that have a an atmosphere of living dwelling um there a sort of livingness to an environment so I'm interesting these subtleties and

then amplifying those subtleties in architecture the idea of intimacy and privacy in the public domain Nooks of a doorway entry or this um I guess almost cinematic memory uh that's embedded into cities um the organic growth of cities the bricolage of cities how growth happens with different personalities and how they form organic constructs or basic symbolic archetypal classical City terminology um like here in pizza in it in Italy um and also the organic um scaled midcal cities that you see in Europe that have Consolidated that intimate relationship to space solid and void and creating um spatial conru constructs where you have external rooms and internal rooms and the idea of those spaces aggregating social activity so all of those things are overall Fascinations and I put them in because they are just long-term fascinations which are always being recorded it's a kind of like a background noise but so the origins of the work um go back to Sak I think when I was studying in Sak in Switzerland in America um where I think I was fascinated by the idea of coexisting realities in structures so I call this the hyper realism of Life reality okay so this for instance is a sculptural piece that I did um about um creating a new construction out of an old construction so uh looking at how an old building's column and beam structure was working and how um The Association of the original thought of that structure which was used the drawings if you like the drawings were the perfect model of that structure then you have the actualized construction which is slightly different than the idealized one and then you have maybe the course of time and organic processes changing that structure and then of course you have your your memory and projection uh that's that's pushed onto that when you experience it so this idea of having an assemblage of all of those realities made material into a structure I became fascinated by that in order to form a trer basis for architecture so that's kind of like back in um the early 90s and then that sort of morphed into looking at architecture more like a body so looking at the more um organic substrate of architecture being much s more similar to a body um and looking at how a molding process around the body and create a replicatable architecture so sort of in a sense A Primitive version of a data fed architectural model being built as a modular system but replicating more of a kind of bodily expression so forming a very intimate relationship between body architecture and making a very ambiguous um relationship there rather than a a sort of um inside outside or dualistic relationship this is much more blurred um so that that I guess indicates maybe an artistic and curious mind being played out as a young architect and then fast forward that into reality into professional work um I started to work um back in 20072 2008 in in healthcare projects and this was one of my first projects um uh it was it's for the most acute um Hair Center for mental health patients in the kind of lower socioeconomic

District of Melbourne called dandong and what we assembled here is an idea of deinstitutionalizing a mental health hospital by using the house as the model for the building and assembling essentially those was like about 30 houses around different Courtyard spaces and making everything out of primarily recycled Timber and then changing the relationship between the building and the people inside but also the people inside the building and the public so deinstitutionalizing all of those relationships and normalizing those sorts of things where people would probably walk past the building and think it was a health Spar or some I don't know Health hotel but in fact it's actually treating the most acutely in need patients inside and then using the courtyards as a kind of natural oasis to feed positive energy into those people okay so there's that these buildings are um are closed in and around Courtyard spaces that focuses the energy courtyard and each Courtyard is designated differently and that sort of relationship is more like a Closter so it's more like a kind of classical Closter monastic Closter where peace of mind and mindfulness comes about through that focus and on that singular um place of peace right so we using here materiality um nature that's somehow getting in between the building and the footpath and then the focus of nature on the inside to re assign the relationships between those people's um Minds in and themselves but also their minds in relation to the outside world so these are how these sketches and ideas got translated into spaces that change the more hard-edged relationship of buildings to Nature uh private realm to public realm into much more of a harmonized uh fluid sequential picturesque sequences of space where we're able to bring in a more what I would call a soft Fascination rather than a hard fascination reality for those people inside the building um so you can see for instance the almost beach-like environment around the building that that forms the border the framing of the building before you get to the actual public realm so it's a kind of like an imposition of raw nature raw Bush into the public realm that helps to harmonize these people and and another project that came about slightly later was a very um prestigious physicist in in Melbourne who needed to have a domain where he could work and and do his algorithms his uh experimentation and he had a space that was in a very urbanized environment on a on a very harsh Street Edge so we created a cabin uh for him in the city in a shop um so essentially um one had I don't know whether you can see my cursor but there was a sort of Street on the bottom edge of this image this is the plan of how we built it um there was a street on the bottom edge with a lot of passing traffic and people and and in order to kind of recalibrate the space so that it worked to generate positive regenerative um environment for him we entered from the back and we arranged a sequence of Entry spaces that essentially blackbox

experiences that would um kind of launder the urban fog or the urban pain of the city the hard fascination of the city launder that out of his body before coming into this Timber domain where he would rebuild and Connect into the truth that was the overall kind of existential phenomenological Journey that we were setting so as we walking into the space we were walking through these deep um kind of lost spaces that were all black with a scrunchy kind of um um Floor Covering different Acoustics and then you would go through this door towards the light it would open up to you and you'd be bathed in this kind of warm light where each of the Timber elements was retained its organic [Music] you know wabby saabi esque character and you know it worked such a it worked so well for him he's never kind of um uh I mean he's always been extremely supportive of the fact that I did this for him because he was kind of it proved to me that on a oneto one level if you listen to someone's needs you can then create an environment that has a major effect on them psychologically um okay and then part of the journey and this is part of the kind of rational um trajectory of the work we worked with a very uh preeminent company called BHP bullettin it's the world's largest Mining Company to do their headquarters in Melbourne and in this project we were able to change the relationship between how a corporate office worked in a city and how the relation ship between in a sense an institutionalized corporate entity and the foot path the streetcape and the history could coevolve so that what's typically a very dry and alienating environment could somehow evoke the past the memory of a place and create a different material reality and material associations for everyone who uses the building and this project sat on an old site that used to be a theater um and it sat in the district of the city that used to be a manufacturing District so they used to make fancy Goods like they used to make jewelry and watches and things like this so what we wanted the building to be is some sort of Jewel like project um that spoke about that and and sort of heroz that history we brought that into the space construct that linked the two streets that bounded the site and we created this kind of sequential Journey that detuned people from the busyness into the quietness um through the use of natural materials and you can see here at the back of the site how the warehouse District gets mapped into the structure and embedded in a slightly different way using cut glass to make it more ceremonial and exotic and that's how the project kind of sits in the city and you can see the Echoes of the relationship between the gothic uh there's a river also sits down here so there's somehow a relationship between the environment the gothic the river and the past all sort of assembled into an architectural approach that's specific to that side um in terms of residential work um we also been working on um some very interesting um Edge conditions in the city of Melbourne where we've been

able to explore the the tall building um typology and how tall buildings can work as Civic buildings um and this is an edge of the city where lots of parades have happened in the past uh edged by Parkland old buildings that were built uh in the sort of mid 1800s using Stone and so what we were keen on doing here is establish an a new architecture that's away from the glass curtain wall driven um way of making things and return the city back to heavy materiality and extruding what would normally be a Palo scale midscale building into a taller building so that essentially your physical uh experience of the public realm was in about continuity but then also the expression of the building on the skyline because cities are seen from afar our city here is a big city in a very flat on it's one of the world's largest Bassel planes so it sort of erupts out of that Bassel plane so buildings can seem can be seen from afar we wanted to Signal what the city was about from a distance uh using a different approach and then connected into this sort of edge condition of the city so you're very much taking part in the activation of streets and the definition of Street edges and seeing architecture more as a passive um tool for containing space and defining Edge um but then also uh creating a Nooks uh in this case kind of cave like Nooks around the edge of the building where you feel the weight of the city around you this is a photograph of how that edge condition works um so embodiment I perhaps go back to this embodiment is more about uh and I guess what I'm trying to establish here is sometimes you're trying to rationalize from memory and from a kind of fact based almost like a simple trajectory to create a building other times you're trying to capture an emotion that's perhaps more complex and evasive um so I call this process more embodiment and this is um process that we use for the new Australian Embassy which is just being finished in Washington DC just about to be opened where we were looking at how material like copper can establish itself as the material to denote a country and the experiences of that material in the city of Washington which has a very um sort of ceremonial and hard to grasp institutional reality and to kind of insert something that was more open deinstitutionalized welcoming and also textured in such a way that it drew or created meaning for people outside of the building but also as you come into the building what those experiences are where of about Australia so we we're looking very much in drawing experiences in plan form and then using the copper to evoke this mysterious uh acology of of Australia that's particularly evident in the ephemeral uh Salt Lakes that you see in the desert regions so if you ever fly over Australia it's just the most beguiling experience that you'll never forget and it was trying to establish a dialogue with that for people in Washington um that's a picture of on top of all Ruru and you get to see the kind of monumental modular

monumentality of that and we wanted to establish something that was kind of rock like or big and bulky and solid and then insert these fishes into it and these are some of those sketches so this is again um it's interesting kind of working remote from another city and inserting a piece of a country's DNA into another city but here we're kind of looking at the dimension the the dimensional reality of the building and how texture can get worked into a building from outside to inside and how nature can establish itself in different ways different experiences so breaking down the form of the building and then we started to look at because we had to What machines and Technology would be able to produce the material that we wanted so we actually get into uh how the heating of the copper would work and how we would create kind of organic replication so a a process of making material that has a lot of idiosyncrasy associated with it but that's replicatable so we use this heating machine a top- down heating machine which the um material will be rolled through and it established this polychromatic kind of burnt finish uh within which we had a just an infinite amount of variegation very beautiful so this whole process evolved around the the making of the building around using that polychromatic finish to create um an echo of Australia on the outside of the building that would stay silent but beguiling um and then the way that those different panels were used was to create a kind of inflection and that inflection that we had on the facade was coming through from The rhythms of the forests that interspersed in the desert and that was kind of mapped onto these sequences onto the building which then became kind of fractionalized and more granular the closer you got to the building and we would study that in virtual reality now because we're able to and you're able to add the environment around it so you're able to choose your charm of day all those finishes and you have a much more closed loop reality when you're designing now that can get very much close to the finished position the finished state in advance so you're able to have a much more exacting feedback loop design system to embed those feelings and emotions into a structure and get it right so this is the kind of flickering and ambiguous layering that I was trying to talk about earlier in an natural Forest uh eucalypt Forest in the High Country and then that would come into the workplace environment through a sequence of um there's two atriums that are collocated one's a public Atrium one's a private Atrium the private Atrium is a hub for all of the different disciplines in in an embassy to come together and so what we're trying to create here is um it's all made out of Timber and that Timber is being used to infuse the environment with with an association with Australia in particular the light the striated light and the raw Timber working together and the spatial sequence being kind of open and expansive and tall like a eucalypt tree so eucalypt tree is super

tall has a big open canopy and it has this small Leaf kind of delicate light that comes through it so all of those feelings we're trying to trap into a workplace setting that we know both symbolizes Australia but also is healthy for people inside the building so this is about healthy workplace philosophy being used as a ceremonial gateway to Australia and these are the further images of that and that's has its currently stands just about to be opened and that's the copper um standing five blocks north of the White House um as a sort of um edifice of Australia in Washington um so as I guess we were developing um a human focused approach we also you can't escape your responsibilities to the environment planetar speaking so we also worked with aspects of the circular economy and looking at how um a temporary building can can utilize wood in the form of paper so we created an artwork stroke building a temporary museum for the National Gallery of Victoria in a competition entry which um looked at making a Timber modular Gallery with paper rooms running through it so the paper which is obviously traditionally been used as a as a an element that you would draw on or paint on but in this case the paper is used as a building material so the building material itself is or the space itself is the canvas and because of its translucent nature we were looking at how the environment around the building would map on to those spaces so in a sense what you're walking into is a spatial immersive spatial experience of the environment moving around the room through the you know the sun's movements mapping different realities whether it be trees or the framing of the building into that room and for that environment to uh change over time so with the rain and the Sun the actual degradation of that environment would also be present in the environment and then also we were able to infuse the um we used to use primitive days in Australia the glue that comes out of eucalyp trees to combine with Ash to create a kind of cement so the idea of using those different natural glues to create smells ukp smell for instance is used as an antiseptic um you can use um a paper uh material like a kind of sponge that can emit um smells and and um almost create a kind of natural outdoor environment inside you know what I mean so we we were interested in developing those naturalized experiences in cubic spaces um and then basically collapsing the whole building and recycling it making it into a book that was the whole concept so it's a popup building that was made into eventually a book we've also been working in um in Asia this is a project we did in South Korea two years ago and this is a um an embodiment project where we essentially understood this is in Soul um that the city of soul is like 80% of South Korea is made out of granite uh mountains so it's a very mountainous country but soul is in a kind of Delta so but it's surrounded by these Granite monolithic Hills and the hills are used to to for people to

escape the city so they go up the hill the mountain and they see a different future they see a different life they get connected to Nature and then they go back into the urban realm so what this project was about is encapsulating that natural getaway type space but bringing it into the city to change the dialogue about what a city could be in the future so we we would let's say decode those sorts of things and apply it to personality types whether it be a company personality type or an individual we decoded process of how those mountains came into being and how the um geometry of those kind of geod like um I think they're called tour structures how they would have they're formed and how they've been ceremonial lied and celebrated through Korean art through Asian art you know the whole idiosyncrasy of a piece of beautiful piece of stone or piece of wood and also it's I didn't know this at the time but seol is also the world center for these um Neolithic structures which in fact uh were hollowed out and used as primitive buildings primitive um um ceremonial places let's call them um so this relationship between the landscape and the identity the cultural identity of South Korea as seen through this ge geological construct being made into an office building that provided this health benefit so how did those Health experiences work on a hillside walking up and how could we translate those experiences into a building so we we having absorbed all of that we would then go into sketch mode and track that geometry and communicate that geometry into plans and into elevations and then into experiences of how you would approach a building what does it look like at night as well as daytime how does the ground plane car parking roadway work with the Subterranean retail how does the how do all these relationships work if you follow that geometry to create a holistic kind of social Tower construct uh that's kind of living version of the mountain brought into the City and you know in terms of breakout spaces um this especially relevant today when we have a question mark about work and about what an office building is um what we're trying to say here that an office building isn't focused necessarily about work and it's certainly not home it's something else and the something else thing is this sort of natural experience or place of elation place of Joy where you go to to get enlightened so the idea of the building being a kind of tool for enlightenment to generate better ideas more Progressive ideas and using nature to achieve that those links so that's what all this is about it's about sort of well in some sense it's sentimental but it's it's TR having gone through the process of embodying those physical experiences on site trips that we did and photographing that and studying the cultural identity it's then allowing us to then create that Matrix of experiences into a form which go top to bottom across the building and form a different dialogue with the public realm so very much a building that changes throughout the day and

forms different dialogue now in terms of formulas and I think formulas are very relevant topic the first Formula that we put together was for the new Royal Children's Hospital here in Melbourne 2007 which was opened by the queen where instead of um let's say forming a kind um a a sort of what I call it an artificial uh Accord with the children who were being um dealt with within the within the environment of the roal Children's Hospital we saw them really as small adults as opposed to children so we saw them as sophisticated uh people just like adults so so so the first issue was to not create a language for the building that was going to be somehow derogatory or form a a poor a poor support structure a poor support scaffold for the children so we wanted a sophisticated structure that could again elaborate that internal world of the child um think about how illness Works in children who are stuck in their beds for say a month bedbound and institutionalized and and that sort of fearful relationship between doctor patient patient in big billion dollar Hospital change all of those relationships so that it's no longer an institution but a place where you again not at home you're not in a hospital you're somewhere else it's you're in a place of Enlightenment and it's that enlight M which improves your health and restores your energies so seeing children differently than what we're told to think of children as being these sophisticated creatures that are curious about the world and assembling cognitive cognitively assembling their world every day just like we are and forming identities in the same way so we sort of created a what might be called the first health code salutogenic health code for hospital and we didn't just use the salutogenic concepts that came from Aaron antonovsky which are all about coherence we B in um concepts of evidence-based design which stem from Roger ol's work on natural light and then looking onto nature the kaplan's work on self festination and nature and their concepts of attention restoration Theory and we we looked into Reg schools in Italy and we perhaps added another two or three concepts on top of that and we used the assemblage of Blended formulas to create the hospital so it's not strictly about either evidence-based design or salutogenic it's about an assembly or diversity um but essentially we're creating an environment that was made into an adaptable building but the environment was the was the driver and the en environment was there to form an escape from habitual activities and that would form a kind of soft interface a soft and reflective interface that would retain your attention effortlessly so this is all about kind of capturing a quality in a building that is very similar to the qualities that you would have walking through a park where you're continually drawn into detailed areas and where you see the movement of light and air and the general ecology as a living environment surrounding you um and washing through you so the idea of having a kind of soft washing of a soft

environment how do you go about making a hospital out of that so we systematize that this is just a diagram of how the salutogenic process works in terms of building coherence from the the focus on building intimate spaces as well as public spaces as well as family spaces changing the relationship between how care goes from being you going to the care to the care going to you so the model of care here is about doctors and nurses and the Health Care system going to the child not the other way around um so it's it's complex in of it self and that was a major tool for how we designed the building but then we're always cognizant the natural nourishment that we get from natural light natural air currents uh pointing the for instance this the building towards the Sun so we embedded the Cadian Rhythm into the building and essentially made a building that was very similar to Forest therapy um by mapping the journey through a path and translating that into a building we would naturally have a good environment for people to be in so so this is the first time that we're really starting to focus on the pathways between the mind and the environment and how it changes hormonal uh structures or how it affects our central nervous system Etc to produce positivistic results so this kind of linking between the environment and the body the mind and the body and architecture for the first time was a established for us so in terms of intuition the idea was to we had an urban edge condition but a fabulous Park that we designed the building into so we what we were seeking to do is to push the building and migrate the building into the park migrate Urban form to point it towards the Sun so that the the building ended up pointing due north and by doing that we're able to create um um Parkland around the majority of the building um and then what we were doing is embedding through drawing experiences of that parking setting onto the outside of the building and through experiences of entering and walking through the building and then focusing in on the child in the bed and what their own experiences of selft Fascination could be within that environment to make sure they're restored as best as possible so we created effectively a model of salutogenesis into a building this is a \$1.2 billion project it had two and a half thousand rooms inside this building and we were dealing with all ages because it had researchers you know doctors nurses researches there all ages focused around children who would start their journey within their old Children's Hospital as babies and perhaps still be going there as adults so we had to deal with all of the age groups in the building this is just a focus on how the night and daytime use of the building Works in relation to Nature the celebration the historical cultural celebration of nature in it's kind of picturesque Australia naturally picturesque um it's like a country garden and using those features in the building this is a picture of that picturesque landscape in Victoria this is

our site uh before it was built on it used to be a cricket pitch and the cricket pitch points due north interestingly points due north and that became where we put our street and we created a street that would form the the Central Social connection device through building um so this was us mapping uh qualities visceral uh textural uh warm and inviting qualities of the park before we designed so we're studying the park studying how the layers of soft fascination work how Shadows work what's the sort of geometry of a tree what's the that that sort of strange ergonomic quality of the trees how they cluster together what sort of light comes through a tree how does a bark texture work and what's the toneal quality of that tree so all of this was documented before we started to design and then we were embedding those experiences circumstances uh we developed up a color palette out of that we created um walls that that reflected the kind of layered soft fascination light quality of the park we went into great detail this is almost like a onetone study of an edge of glass and how dots and patterns and soft curves were being used to evoke that and then we created a bunch of experiences throughout all of the two and a half thousand rooms that evoke that those experiences into architecture principally on the basis that the relationship between you as a person looking out into nature was a healthy one and nature was brought to life through the Sun and what we're essentially doing is trapping the Sun nature energy and bringing that into the building that was the essential relationship and then we were personalizing it so avoiding institutionalized replication we were interested in having each of the two and a half thousand rooms different to one another as well which is pretty high order of complexity uh this is the plan of the impatient unit which was shaped as an organic star shape uh for practical functional reasons but because we did that we're able to break the experiences of the nurse station relationships with the children into smaller packets of space uh so and then we were able to open up the corridors into Collective space rather than Corridor space to make more of a home and a kind of focus point for for the different spaces so I guess what I'm trying to say without going into great detail is that within each of the spaces either at a very microscopic detailed material matter perspective or spatially speaking how rooms relate to corridors or how a building relates to a park all of those the different scales they were all treated with the same level of care um it's again focus is on um again associated with soft desination how this kind of dot matrix that was encircling the building how it resonated with the eyes and how it would change with time of day that would come to life we used energy saving techniques to surround the building with these glazed Sun Canopies were that were made out of colored glass made out of dots of different colors and different shapes and because we're able to map the trees

outside into the space the and because we're making the Sun canopies out of glass we're able to create a kind of transcendent experience of the edge of the building that was bringing that energy deep into the heart of where the children were living and this is how the external building came about so we're literally translating the foreground of Nature and wrapping it into the building to form a blurred relationship between building and Landscape so it's very firmly embedded into the landscape it's an extrapolation of the landscape so in terms of the geometry um it started with the tree geometry and you can see this in the shadows but then if you look at those kind of arms that are coming out of the tree we wanted to utilize that geometry for um how we constructed the larger spaces whether it be waiting spaces or the bridges that that or the textures u in the internal cladding of how these um yeah bridges that that that came in and out of the different fun functional areas of the building would crisscross the street and how they would assemble themselves into these kind of Nooks or torsion sculptures that were very similar to the outside world so whether it's an area of compression or a concept about the scale of an object like the scale of a bridge or the scale of a of a of a staircase how those large scale pieces would relate to a CH child's Mind's Eye in even though you're in a large space these large scale pieces were being assembled to kind of deinstitutionalized but also form a sympathetic relationship to a to to the scale of a child and the joy of a child's Mind's Eye and on the outside just looking at how these sun shades again would replicate um pattern language of the trees into a kind of foliage that changed color and patina depending on the time of day so that was kind of very interesting process and it after I guess we finished it it um it changed my Paradigm or changed my thinking of of architecture as being a tool for health full start so since then since 2007 all of the work that I've been doing has been Associated health and seeing cities as generators for health um and so once you once you get to that position you then start to think well is nature or our job in translating nature all associated with the natural world like a royal park and using software ation or is it something that's more associated with human psychology and the nature of Being Human and I guess the latter concept uh came into being kind of more recently with uh work that we're again doing in South Korea which we called trans buildings which are less simple translations of nature which actually focus more on human nature in buildings and this project was in Ean and it was a project for a a private members club and you would probably know Soho house and uh buildings of such like in big cities where members come together in private Realms to celebrate being flate members um what we were doing here is changing that Paradigm and forming a private members club that was much more public um so unpacking a singular

building and making into a kind of Civic domain so creating public space out of a private building and then in a in the site forming a different relationship between buildings in their environment um making a more ambiguous relationship um so the buildings were seen as sculptures and kind of like non-b buildings um they were they were broken down into pieces so it became a district as opposed to a building and what we were trying to do again is to change the psychological relationship between the people inside the building and the building itself but also the members of the public who are looking onto the building and forming a questioning relationship so kind of eming a an existential crisis if you know mean so seeing a building as being a partner in the Journey of existence in the crisis of existence and the questioning of existence um and not necessarily being a Utopia but being kind of just a living piece of city and we're also interested in how the new world of virtual reality is coming into being so trying to establish a building that can go into the metaverse as well so we collocated the idea of a new virtual building with an existing building to form a more hyperreal um form of architecture which I'll come on to perhaps later in the conversation but the environment was a very interesting uh context where we had the most International District of Soul a place uh fraught with social uh friction um a disheveled environment that was very popular came to life at night so a lot of kind of non-livable features in in very broad sense um that naturally come about in cities and embracing that rather than critiquing it but trying to form very beautiful cinematic environments which were highly evocative making buildings that were charged with a kind of psychological tension using materials that were hard to describe so in this case it was 3D printed wood and seeing the building as an face but having an in a completely different internal world so sort of in a sense evoking what we're like as people we have an external appearance but we have a highly complex and changing internal world forming similar relationships in a building and then making a building that was ambiguous as to what it looked like having gone through um an interest in in uh this essay by JG Ballard his his fabulous uh written piece on the garden of time we then started to don't have time to go into that but we then got into translating all of that into art pieces which then came into physical experiences of the buildings um disappearing underground and the exploration of an underground world so essentially arriving at an assemblage of buildings half of which are above ground half which are below ground appearing as these concrete simple Prismatic volumes and quite Stern and monolithic and using water to at the base of the building to cleanse the buildings and the environment to then looking at how the internal World worked as a more organic place and contrary to the Prismatic volumes on the outside looking at a much more bodily like sequence

of spaces on inside so whether you're arriving at the retail or the the rooms within the private club or arriving by car you're sort of arriving inside a body and having a very different relationship and then the Two Worlds then connected and we started to look at how Windows worked in the building how the concrete would reduce to fabric would would transcend and change into glass and how different relationships were working through different spaces and how that would become organically expressed into different sorts of thresholds which were more more body type thresholds to form a building that was kind of half building half body and then looking at how the concrete in the evening would then recede back to become a more translucent material so the idea of the buildings kind of being more translucent during the night time and solid during the daytime and having that ephemeral quality and now lastly uh we've been working on the new hydrogen economy in South Australia in a place called W looking how to use new materials like green steel how to repurpose industrial buildings but looking at how principally ecosystem how the industrial ecosystem uh can create a more more humanized uh Urban world so we've got we're on the edge here of looking at how the urban reality of this new city that's going to grow because of hydrogen energy how the relationship between energy the Earth manufacturing and urbanization triangulate into a more holistic Paradigm by looking at all of the different ecosystems being embedded into one another so starting with nature and the natural reality of the site the Ecology of the site um how that the ecology has been manifest into structures whether it be Timber Old Timber peers or old existing buildings into a kind of embedded memory of of the site so we're recording in a sense what the memory the cultural ecological memory is of the site and then We're translating that into to a shaped language an experiential language that then overlays into the new housing project that we're doing to generate experiences that capture those that that sort of existing experience and manifesting into a bolder bigger identity for the city so these are sketches as they translate into new house so sort of balance between the sculptural world of found objects on a beach made into to a piece of architecture um forming a new city out of all of those embedded memories so so I guess Daniel and and the audience if we've got any audience left all that um what what all of that constitutes is a kind of flow and the reason why I presented to you like that is it represents a kind of intellectual theoretical practical artistic flow that represents how creative cognition works and I guess the overall ambition is that because of that coiling process we end up with a much more coherent and friendly human celebratory world and that's a world that we all need and deserve wow I guess I would have said I'm not sure where the spiral ends but of course it doesn't end yeah it doesn't end absolutely yeah well

please continue and well I I mean from my perspective I've I see the institutionalization of the world of architecture the world of property development the world of law the world of Politics the world of Industry um having behaved poorly let's say in the past or perhaps narcissistically speaking um evolving systems that serve their own ends in their own ways but we're now sort of in an environment where we can actually bring all of those pieces together and to form a self-reinforcing self-re improving you know Networks and thoughts and processes that can actually inform and and and manifest a better physical world so it does mean that as a artist architect you are interested in politics you are interested in Industry you are interested in AI for instance um because you're interested in consilience you're interested in tying together the different disciplines just like and how Works you're in dialogue with the environment so in a sense all of this is is naive in the sense that all that is is a replication of natural process it was an incredible presentation and I have many pages of notes but first to kind of lead into that I would love to hear about your engagements and encounters with Carl friston at all um well like here um I went through a similar um kind of presentation and I guess what was interesting I didn't know how it would resonate or whether it would whether it was just me fantasizing that it had something to do with free energy and active inference but I had a very strong hunch that it did and I guess what I'm what I got from it that it very very much indeed does correlate very strongly with the embedded principles that Carl's works all about and um I think he was very pleased uh to see it being translated in this discipline in this way um and I felt relieved to be to be quite honest you Daniel talking to someone of the statring caliber of someone someone like Carl um to be able to talk directly with him and and the people around him who are all excellent um was was was extremely fascinating and beautiful um I mean I see it as a kind of the the way I I just see it as the way forward um and I'm excited just to understand how it can translate into into a process in a way not in able to encapsulate the complexity logically so that people can understand can get on board so we can de institutionalize this this logic and spread the word if you know what I mean um spread it around and encourage participation and um so I got I got the kind of if if you're not to mean the collapse of boundaries in the in the conversation with Carl because because it can go into all areas of course it goes into all areas I'm just showing how it translates in the way I do it but um I think it it needs a long time to kind of just sit there and and discussions to to to to Spring from here you know to Spring to get going um I see this spinning into all sorts of different areas so I mean at the moment we're looking at how to utilize this in coding terms so I'm interested in computation and statistics I'm interested in urbanism like making new cities um

so how we translate this into we're building a \$200 billion new rail loop around the city of Melbourne how this can translate into new districts and new cities and how we can break down this typical artificial boundary between new things and I don't know new cities which are typically unfriendly unloved and unwanted into being coveted people aspiring to move move into these things right and we never got that right ever um I mean since Paris you know we've never got it right so the idea is to actually is to get something finally up and running which represents an opportunity to to do something at a regional level a local cultural Community level that per perpetrates Evolution at different scales in different areas through a process kind of Open Source very fascinating again there's there's so many places that I want to just touch on because I I think you do present so many contact points and ways of expressing that will resonate with active inference enthusiasts and yeah I I guess the starting point is the learn dwell Nook so you were taking those measurements because they were informative you were updating your cognitive model from your sensory input and that changed your generative model it also may have been actionable information it may have been folded through an iterated realized process back into the design and then fundamentally that's an attracting space it's an attractor or a dwell space just like if you had a landscape and you were shaking up the a ball on the landscape it would dwell as you turn down the temperature and slow down the shaking it would dwell probabilistically according to the features of that landscape and that concept of attractor points attracting sets and then also wandering sets and moving yes Nooks which is something that may be able to happen in a digital space or it might be a mobile element or movement is all just a question of over what time frame that idea of the attracting set and of things exist by virtue of what their attracting set is that's the heart of the free energy principle so starting with the dwell on a surface and then I'm sure there must be a point in the design where it's like so we saw 50% this and 50% that do we want to carry that forward or are we looking for like 8020 or do we want 40 4020 with a new affordance being introduced through through landscape modification it completely fluently um touches on it and I just love that learned well photography because I think probably many of us with cameras in hand now have our own little peculiar brand of Photography and then what does that say about us yes what Daniel what do you think about this whole um world that that s in terms of free energy active inference and how cognition works in relation to creative process creative cognitive process as let's say creative cognitive process being a kind of symbolic representation of cognitive process as a sort of exploration as a projection what do you think about that well we may have to revisit it multiple times but here's what I wrote down perhaps

touching on it you showed a image of of a photograph of some metal that reflected the idealized and the actualized construction of the ibeam coming together yes and that both of them had differences that were uh materialized differences through time and weathering and context and it was an assemblage so there was like all of this richness and I thought wow it's an ibeam that's like the self that's like the eye and then there's a comparison of the idealized eye even of just a proximately idealized eye like I could be standing up right now or I could pick up that object that kind of idealized counterfactual adjacent possible the abductive the associative the creative that exists in our lional proximity of the eye and yeah and then it's all about that what the image that was shown with this sometimes horrific but sometimes beautiful ju deposition of the real and the actual eye or moving it a little bit out of psychology like the real and the actual self model and then our calibrator on that is variational free energy yeah but you know this kind of how do you say it um uh the concept of the the talk was about naive architecture or naive approach which is a kind of you as vessel for that complexity so you're sort of living manifestation of that complexity as a person person as a human being you're not you know I'm not a tree I'm not a plant I'm a human being but that cognitive process and natural biology and The Coincidence of those two realities being brought together as a cognitive process that manifests that expression of the well it's the it's sort of the first Loop is you absorbing and then expressing and then the expression becomes the physical manifestation of that thought process being the substrate that you live in right so that whole chur chilian concept of you know buildings make you know we make buildings building make us that whole loop system um you know it's obviously been been proven to be to be right and cities can either be pathogenic or sologenic you know how many cities in today's world in modern sense are salutogenic cities none you know you go you'd have to go back to the Villages and towns and blah blah blah which which are very much um tight manifestations of humanity in in a sense in a physical manifestation sense um and the abstracted I would call it Excel spreadsheet logic where you take one element and then extrapolate cities out of that one element loses all of that Nuance so now being in a situation where we understand that nuance and we understand the importance of that Nuance there are elements but also links in that Nuance Matrix that we can almost calibrate and we can we we can capture them we can embody them we can recreate them and now we've got this kind of opportunity within that sort of febr Matrix of stuff that's in part facts that are math-based facts in part just visual U Reflections or actual photographs um but we're now able to kind of like bring all of that together as a sort of more complex arrangement of things to manifest something as the new

paradigm um that forms that sort of partner in in our cognitive process uh that's not an enemy to our cognitive process that forms the coherence and the um natural it's our natural place if you're not mean it's the sort of hiiga space it's the place of dwelling yeah a lot of angles it's the William Blake dialectic Songs of Innocence songs of experience but I want to connect again the visual representation what what do these visual what what even are we looking at it's something I asked myself probably on 2third of the slides I was like is this if there wasn't a person in the frame I was like yeah is this a microscope is it VR is this aerial footage and sometimes probably even if then later on oh that's four trees but it's like but is it and all of that richness of of the the temporally thick interpretive process and also the relationship between like the synchronous Audio Visual and the I secades a lot of other features that we can quantify um there was a scale freeness or a scale ambiguity a system ambiguity and the mathematics of active inference and the free energy principle are scale free or we could SK say scale friendly which is to say they don't host or hinge on any specific or particular system or scale it's not a framework about brains but it can be used to paint or Express models of brains and so that meta theoretical position which may have more resonance with the creative encounter or the kind of like massively collaborative and Global projects that you described that setting from a practice angle especially with augmented and synthetic intelligence may come to be closer to the process of active inference modeling than 1995 or 2015 writing a statistics program yeah I think so I I'm with you on that I think it is I think it's really it's kind of experimental actual experimentation of of the free energy the world of free energy and the successes of it we can see and experience um that's the joy of it and I think I also would like to say that with studying how creative process works you're almost being inside free energy as well I mean I this is just me talking and I don't know teally speaking whether I'm right on this or not but it just feels to me that it's a rich source of um information to unpack that actually could provide some really good strong direction for the future the the piece that really left an impression on me that makes me think you you said right there the successes the realiz successes successes only in principle or hypothetical the real ones are real and you mentioned in the copper for the Washington DC building that you wanted to make it idiosyncratic yet reproducible and that's like the magic quadrant like yeah we have Technologies of reproducibility without idiosyn cricity plus externality Plus aect um path pathology Etc and so this quadrant or space with the idiosyncratic and reproducible and then it was like it's a top- down heating machine and so in some ways the realized artifact becomes like The Crucible or the eye of the needle where the generative process the process that leads to the artifact's deposition um or accion or

however it's built it's a demonstration of the sense making and the Aesthetics and the technical capacity it's like wow they brought that rock from here to here and then yeah did they do it this way or that way and people today discuss and debate how did they do that building how did they do that building and so it's like the archetypal Logistics operational setting construction physical Construction ruction is the tip of the iceberg of the physical and informational supply chain and all of those for those who have the generative models to see we look at the Obelisk from Rome and that is like a big rock and it ripples for those who have knowledge of that setting but also the footprints of an animal to another animal or the phermone of one ant nest made to another ant mate or all these other like infinite last miles and those are all yes ANS with each other they're not in contradiction to each other and neither is it a contradiction to have a unified way to discuss and approach that yeah so so within your work and within the studies of act active inference how are you critiquing or unpacking modernity in cities in buildings how is that being how is the world of active inference being explored in the physical domain or is it or is it being explored or but when it is being explored how sophisticated or or longterm has the research been all great questions the first thing that came to mind is that there's been some work on the affective and the physiological and the interceptive and while there has been for example models of movement within a maze to really get to the point of a a simulated model being more than just an intuition pump which can be more than enough itself but for it to be more than that really at this point in time requires a a team professional effort that to the few extents it is happening is probably happening in proprietary settings but we do everything we can to highlight the open source capacities and the kinds of models that people are building I think one setting that is pointed to a lot but also has not received that much empirical or open-source work is like like ecosystems of shared intelligence instead of just talking about the physicality of the internet of things or iot or everything in that space being able to integrate the spatial and the ergonomic like you pointed to earlier with efforts to provide something like a meter stick but we're not just looking for distance we want something to calibrate the construction or the architecture process and at hand was the Industrial or potentially modern system and then what I saw right from the beginning of the presentation was like a yearning for a compliment or a juxtapose or the shadow of that and affective cognitive science now being able to kind of square that Circle or however you want to think about that where it's like well we could organize our project around people's sense of narrative fulfillment and that could have disastrous ecological consequence so let's care about the narrative and the ecological okay but then the business might not exist all right bring it into the

generative model what El what do people think about what we have now and that must iterate that must be Intermodal and so that allows the imbu of all these different aspects that that for me were evoked when you talked about like soft attention whereas in science and technology and math which was only onethird of your threepiece pie a soft attention can be a boon for a beginner in scoping at the space it can also be um a pleasure and a tool for an expert who's able to have a soft background and then sharpen attention um but to be useful one has to be on the other side of that innocence and experience and that's a tough learning curve because there's and no one is ever going to be able to have all of the hard attentional regimes there're just too many it's just like trying to know every language or like know it's like it doesn't even make sense it's not even preferable like for any number of reasons and then it's like well so what do we do but it's like but there we are scratching our heads talking about how we're going to create living re and make it work physiologically if not far far more than physiologically a as a group as a team as probably discussions have always happened leading through to the inculturated and lovable structures that we love and so I love I love how you use words spr is fantastic um um see I would sort of say that diversity that one is enabl to get hold of to grab in and bring in to a creative process if you have a porous or agile process allows you to bring anything in think always made me feel as though when a project came about I would always say well there's a million buildings here which one do we want to to build which one do we want to design and that was kind of a fairly alien concept to a lot of people in my industry or in my education certainly people wouldn't W talking those terms because it had always been trapped by some cultural doctrine that one had to be a postmodernist one had to be a deconstructive a regionalist a modernist a this or a that there was always some sort of virtuous um theoretical underpinning to something that was determining what you it were basically institutionalizing what things should be like for for whatever reason for power influence money whatever it is we were trapping a process thing um codifying it and then telling everyone that they had to be in that box whereas what we're talking about now is totally liberated it's totally free for being what it it should be so my yearning for craft my yearning for vernacular architecture my yearning for that hard to describe soft granular environment that I like so much in the old world in inverted commers or or seen in agrarian structures um I can now talk about in terms of how professionals talk to one another how teams are made how um the intangible has now become sort of tangible through the ability of talking about it in terms of free energy um Neuroscience it's enabled us to actually bridge the gap between qualitative and quantitative and that's the joy of it of being

able to go from an artist wondering in the bull rushes feeling utopian but creative and curious manifesting things to really having a quantifiable opinion about something and being able to capture it compartmentalizing it without losing the nuances compartmentalizing it and explaining it to someone and then you know it should be like sort of engaging someone to design your house quite often what would happen you'd end up with arguments with your ARG you'd then something end up with something that you didn't want um You probably went through some sort of mood board analysis or gave him some sort of brief or them some sort of brief um but there's always been this hard boundary between what you want as a person and what someone else is giving to you you know and I I think what this is enabling is a soft um uh sort of navigation or translation or um Arbitrage this sort of soft and giving Arbitrage between you and the environment you and someone else you and a large corporation that enables both parties to come together around a common set of values that enables that negotiation of that Arbitrage to work in everyone's interests so I think what sits on beneath all of this it's translating yet into an architectural placemaking philosophy and place being a kind of maybe closely associated with coherence I would say but anyway the idea of it being a negotiation tool I love that I love the environment as being an expression of fruitful coherent Arbitrage of relationships awesome and synchrony generalized synchrony with the environment not lock step with the environment but synchrony like turn-taking and meaningful dialogue and alignment with our environment our built environment information architectures our social architectures and and I think this is something that hopefully we'll see unfold in so many different ways is like how do we hold the moment like the aesthetic moment the lived um The Humble home environment moment or whatever moment the affective moment outside or or even before we even quantify the affective or the cognitive like the real Primal moment and it's like maybe one day we'll have tools soft enough to touch I don't think we have them today so what is one thing that we can do and maybe not even regret ever looking back something like create a buffer or a container for the vital moment and so it's like let's let's just make sure everyone's at the workshop on time and that this is the way that they like it and and all these features and be very thoughtful and mindful and value aligned up to the moment of The Craft and and then if there's more to say it may be right to say it and impose then and constrain then but if there's nothing more to say after you've like brought everyone to the dance then that would also be fine and so then the challenge is like but you have to make a building like you're being paid for a building and then there has to be that received public or Civic experience that is not like oh we'll watch this two-hour live stream

and check out this or that analytical framework there has to be the real direct and and I mean those are the kinds of knots and Tangles that it's just amazing to hear your your journey and your way around the kind of I call them I call I've just written this down actually a Nook machine I think what what I do is like Nook machines and so I think by helping the World by creating Nook machines we end up generating a more truthful honest coherent Amplified truthful kind of expression of how we want to live with all of the different instincts activities that we do every day right and it's kind of like the enabling of all of those activities whether we want personalized in you know intimate moments or expressive outward moments but the health of a human being being that sort of inner and outer World being being enabled to effortlessly assemble every day U Melodies and that's sort of idea of a building let's say an environment capturing a multitude of different feelings not a singular feeling but a a multitude of layered feelings to create Melodies and so often what happens in symbolic architectures you're really drilling down in the guard rails of a singular emotive reality and you know none of us live in that singular emotive reality and we don't want to live in a single modality like that so that sort of multiplicity of modalities sets that that sort of complex Melody thing and I think that's that's what we're really talking about is through that communal process or that iterative self-improving process which absorbs contrasts and works through just positions we can absorb a country's voice uh a complex singular voice into another person's voice and the beauty is that synchronicity or that moment being distilled which then enables further conversations to happen which is kind of what I think nature does to us naturally it sort of forms that dialogue um and what we're missing out of cities is and the physical environment is a lack of dialogue capacity so this Nook machine thing is really about creating paracity a living paracity that that allows people to be okay to be people you know because we the city in a sense is telling people that they can't be themselves and that can't be good for it can't be good for us you know so I think we're getting close to kind of having a kind of complex modeling process that then can feed from data that we can find that's say to improve the modeling but you know we're on a journey of course but we're at the point in time where we can probably get a lot closer right now in objective terms in rational terms to get that poetic feeling distilled and to be able to tweak it to be able to sort of play with it tweak it synthesize differently sounds and the music coming from it before we build something and then we also can start measuring those qualities after after it's been made and literally feed back that information to the next project wow well in our sort of closing is thoughts a close synonym for Nook is also Niche like I've heard a breakfast Niche and in Ecology of course we

talk about different aspects of the Niche and in cognitive Science 2 the Developmental and the cognitive Niche and so it's like that's the map in the territory the cognitive Niche is where your belief distribution is hanging out and there's a cognitive Niche about the breakfast nook like we ought to have breakfast there or we're the kind of family that uses the breakfast nook for that but it's like whatever it is on the map and whatever it is on the territory with the of course digital being a complexifying factor in all of this but whatever it is let's just go with that and have a path of least action on the generative model active inference with a free energy principle and then we can talk about what we expect and what we prefer like we expect everybody for breakfast at exactly this time versus like we have loose expectations or it could be uniform across the whole day we can talk about our expectations and preferences no appeal to a reward people aren't loitering and lounging in the park for a reward function directly or per and yet that is what the kind of Quantified Matrix and the reinforcement and the reward learning which is not wrong it's just a special case it's a partial constrained expression and a partial constrained regime on all sides of the blanket and so I don't know where it all goes from there but Daniel you know the the learning dwelling Nook concept the conflation of having a synchronous um let's say density of want where you're both intrigued and curious about something but also kind of want to be in it so you're sort of learning you you as you're moving through the environment you're seeking to learn and on top of that you're seeking to gain and arguably seeking to inhabit or dwell better or improve your chance of dwelling better um as some sort of effective strategy for survival yeah I don't know whether that's true but it it's quite actually the two components of expected free energy you said to learn and to gain epistemic value pragmatic value these are the two components of a unified imperative and we want a high reliability setting where it's like you're not here to learn we need this done that way yeah and then there's also the other end of the pool and there's all the Dynamics and movements within and it's like that's that's the pallet now what do we want to say yeah yeah um and what what needs to be said comes out of the the arbitrating process uh which goes into a lot of detail because doing a building for instance takes like seven to 10 years sometimes you know know it's not a short process um and the reality is that they're kind of like codes it's like you're building an algorithm it's just it's a physical algorithm it's a physical embodiment of knowledge and that knowledge creates physical experience that people intuitively experience without you know maybe dwelling on the seriousness of it or the impact of it people are at least experiencing it and so this the trouble that architecture has always had in the modern era is evaluating the power of architecture or the meaning of architecture to

health and wellbeing and the growth of culture why why is architecture important is it important um and if it is important how do we quantify the importance of it um and and Architects and the industry tend to if they're not maybe born into environments where they naturally get it if you're born outside of architecture in a sense maybe you've been surrounded by industrial reality the whole of your entire life you may not I'm I'm just guessing here but you may not fully comprehend at a bodily level what the meaning of that is and why it's important virtuous beneficial I don't know but but for Me intuitively Maybe This goes back to archetypes being an artist architect looking to looking and being curious is is a foundational thing for me at a personal level personality type level but so is the idea of creating structure so the idea of creating on one hand and the idea of structure on the other so creating scaffolds and so um that's my particular personality type um and I feel as though it's almost my job you know it's like I've been born to do a job and I'm enacting that job I'm just trying to unpack what I would naturally urge what I've been naturally urging to do my entire life uh to the point where um if someone says why does this space feel so good it's it's much more complex than just saying well it's got good natural light or or um the Acoustics are good or it functions really well there's the density of the psychological reality that people never get to talk about and there's never really been a framing of that I mean hiig has had a good Bach heiger have had a stabed it um but it hasn't had the sort of um balance um of dealing with highs and lows the different tonality of Life uh the naive to the sophisticated the the poetic to the I don't know the everyday there's never really been something that systematically drawn all of those constituent elements together and it's always become a dry dried up kind of artifact a dried up almost purposeless lacking purpose Theory that sort of begins and ends and then it's just there on a bookshelf for other people to learn about but there's never really been an active philosophy that has translated and improved continuously over time and there's something about neurobiology and the fact that it Bridges bridges into creativity in the physical environment that is the kind of for me it's the um it's it's everything really it's the it's it's the moment this is the moment when it actually comes together and we're able to actually use these kind of Golden Threads and make make some magic and that magic not being some sort of um accident but actually having a tangible realizable teachable or open sourceable kind of reality to it we're ju we're just at that moment I think where we can begin that work thank you would you like to make any closing statements and then I'll make a closing statement um I I think this first of a lot of conversations I just think this is the primer for me it's the primer so watch space thank you you I hope so this was unique in multiple ways and a delightful experience

and and a real journey and I'll close because you invoked a golden string with the quotation that William Blake had on his Tombstone constructed he wrote I give you the end of a golden string only wind it into a ball it will lead you in at Heaven's Gate built in Jerusalem's wall very nice very ni thank you till number 60.2 yes till next time it's been awesome thank you thank you see you bye bye e